Movie Recommender System

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- Live API deployment
- Model versioning + A/B testing
- Monitoring + traceability

SLIDE 2 — System Architecture

```
(.venv) PS C:\Users\rryza\Downloads\MILESTONE-4> docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED        STATUS         PORTS                                                                                                   NAMES
39c63375840f   confluentinc/cp-kafka:7.4.1      "/etc/confluent/dock…"   3 seconds ago   Up 2 seconds   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:29092->29092/tcp, [::]:29092->29092/tcp   milestone-4-kafka-1
b058dd805343   confluentinc/cp-zookeeper:7.4.1   "/etc/confluent/dock…"   3 seconds ago   Up 2 seconds   0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp                                                             milestone-4-zookeeper-1
(.venv) PS C:\Users\rryza\Downloads\MILESTONE-4>
```

- FastAPI backend serving recommendations
- Kafka pipeline for event streaming (local)
- Model registry for version control
- GitHub Actions for automated retraining
- Render deployment for live API

SLIDE 3 — Live System

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> $env:KAFKA_BOOTSTRAP_SERVERS="localhost:9092"
>> $env:TEAM_NAME="team"
>> python -m uvicorn service.app:app --reload --port 8000
INFO:     127.0.0.1:65148 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:64775 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50122 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50122 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50133 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50139 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50133 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50139 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50147 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:50147 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59891 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59900 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59891 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59900 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59908 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59900 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59908 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59914 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59920 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:59930 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:49195 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:49204 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:49212 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:49220 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:49227 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:60392 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:60403 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:60411 - "GET /recommend/2?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:60417 - "GET /recommend/3?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:60426 - "GET /recommend/4?k=5&model=auto HTTP/1.1" 200 OK
INFO:     127.0.0.1:64491 - "GET /switch?version=v20260413235853 HTTP/1.1" 200 OK
INFO:     127.0.0.1:64514 - "GET /healthz HTTP/1.1" 200 OK
INFO:     127.0.0.1:51233 - "GET /recommend/1?k=5&model=auto HTTP/1.1" 200 OK
```

- API deployed on Render
- Handles real-time requests
- Returns recommendations instantly
- Behaves like a production service

SLIDE 4 — Health & Availability

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> curl.exe http://localhost:8000/healthz
{"status":"ok","active_version":"v20260413235853","uptime_seconds":1057.53}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

/healthz endpoint monitors system

- Shows:
- system status
- active model version
- uptime
- Meets ≥70% availability requirement

SLIDE 5 — Recommendations

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> Get-Content artifacts\provenance_example.json
{
  "request_id": "c52c892e-2a2e-469c-a22a-aac9394369b3",
  "user_id": 4,
  "model_served": "popularity",
  "ab_group": "A",
  "model_version": "v20260414000022",
  "data_snapshot_id": "watch_events.csv",
  "pipeline_git_sha": "bbe12d5",
  "container_image_digest": "local-image",
  "recommendations": [
    50,
    172,
    10,
    90,
    300
  ],
  "event_ts": 1776126233.6023877
}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

- /recommend endpoint returns results
- Input:
  - user ID
  - *number of items (k)*
- Output:
  - list of recommended items
  - metadata for each request

SLIDE 6 — A/B Testing

```
"model_served": "popularity",
"ab_group": "A",
```

- Users split into two groups:
  - Group A → popularity model
  - Group B → item-based model
- Enables comparison between models
- Built directly into API logic

SLIDE 7 — Model Switching

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> curl.exe http://localhost:8000/healthz
{"status":"ok","active_version":"v20260413235853","uptime_seconds":1648.64}
```

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> curl.exe "http://localhost:8000/switch?version=v20260414000022"
{"status":"ok","active_version":"v20260414000022"}
```

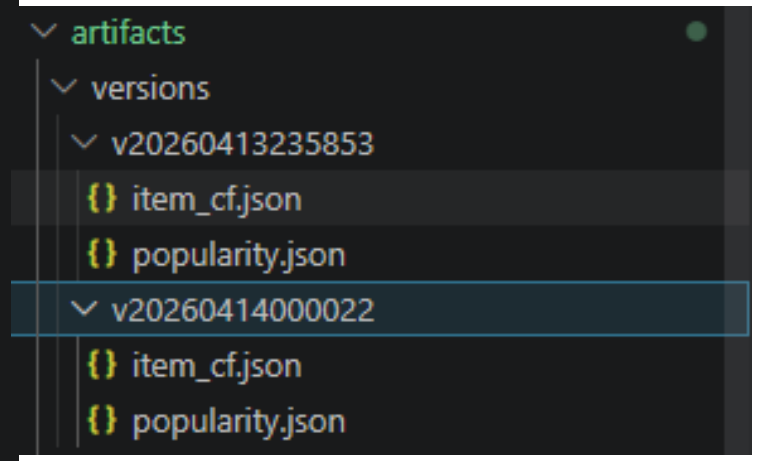
```
PS C:\Users\mdzie\Desktop\MILESTONE-4> curl.exe http://localhost:8000/healthz
{"status":"ok","active_version":"v20260414000022","uptime_seconds":1669.54}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

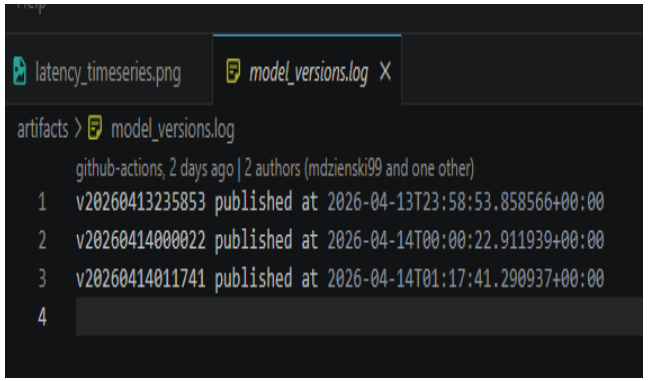
```
PS C:\Users\mdzie\Desktop\MILESTONE-4> curl.exe "http://localhost:8000/recommend/1?k=5&model=auto"
{"request_id":"fd3ec551-7856-477b-b11f-072d099926f5","user_id":1,"k":5,"model_requested":"auto","model_served":"item_cf","ab_group":"B","
model_version":"v20260414000022","data_snapshot_id":"watch_events.csv","pipeline_git_sha":"bbe12d5","container_image_digest":"local-image
","recommendations":[90,300],"event_ts":1776127537.1738356,"source":"api"}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

- /switch?version=... endpoint
- Changes active model instantly
- No restart required
- Supports safe deployment

SLIDE 8 — Model Registry

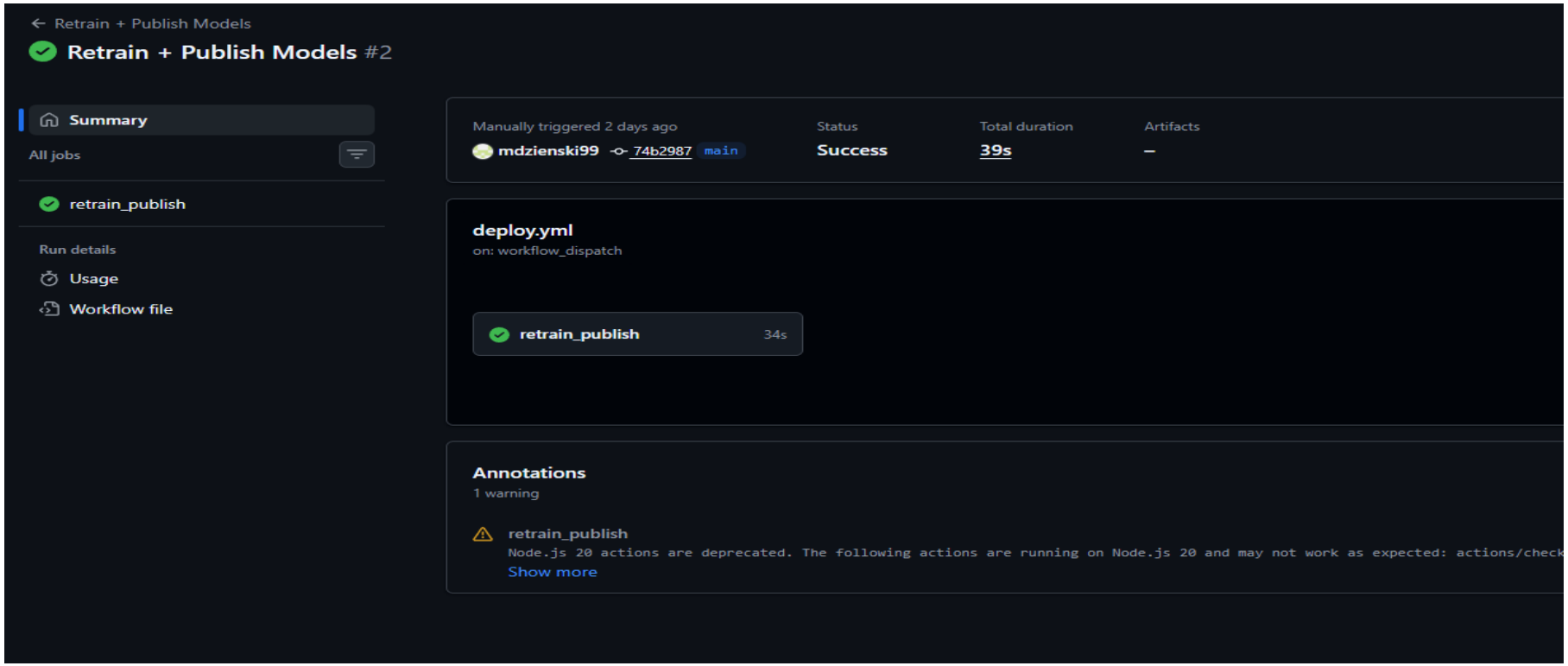
```
PS C:\Users\mdzie\Desktop\MILESTONE-4> Get-Content artifacts\model_registry.json
{
  "active_version": "v20260413235853",
  "versions": {
    "v20260413235853": {
      "created_at": "2026-04-13T23:58:53.826012+00:00",
      "models": {
        "popularity": "artifacts/versions/v20260413235853/popularity.json",
        "item_cf": "artifacts/versions/v20260413235853/item_cf.json"
      },
      "data_snapshot_id": "watch_events.csv",
      "pipeline_git_sha": "bbe12d5",
      "container_image_digest": "local-image"
    },
    "v20260414000022": {
      "created_at": "2026-04-14T00:00:22.888534+00:00",
      "models": {
        "popularity": "artifacts/versions/v20260414000022/popularity.json",
        "item_cf": "artifacts/versions/v20260414000022/item_cf.json"
      },
      "data_snapshot_id": "watch_events.csv",
      "pipeline_git_sha": "bbe12d5",
      "container_image_digest": "local-image"
    }
  }
}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

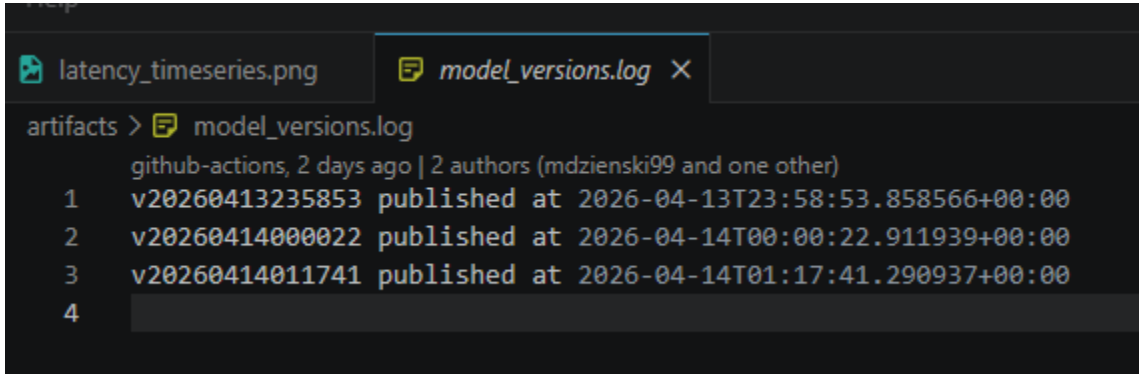




- Stores multiple model versions
- Each version includes:
- trained model files
- metadata
- Enables rollback and comparison

SLIDE 9 — Retraining Pipeline

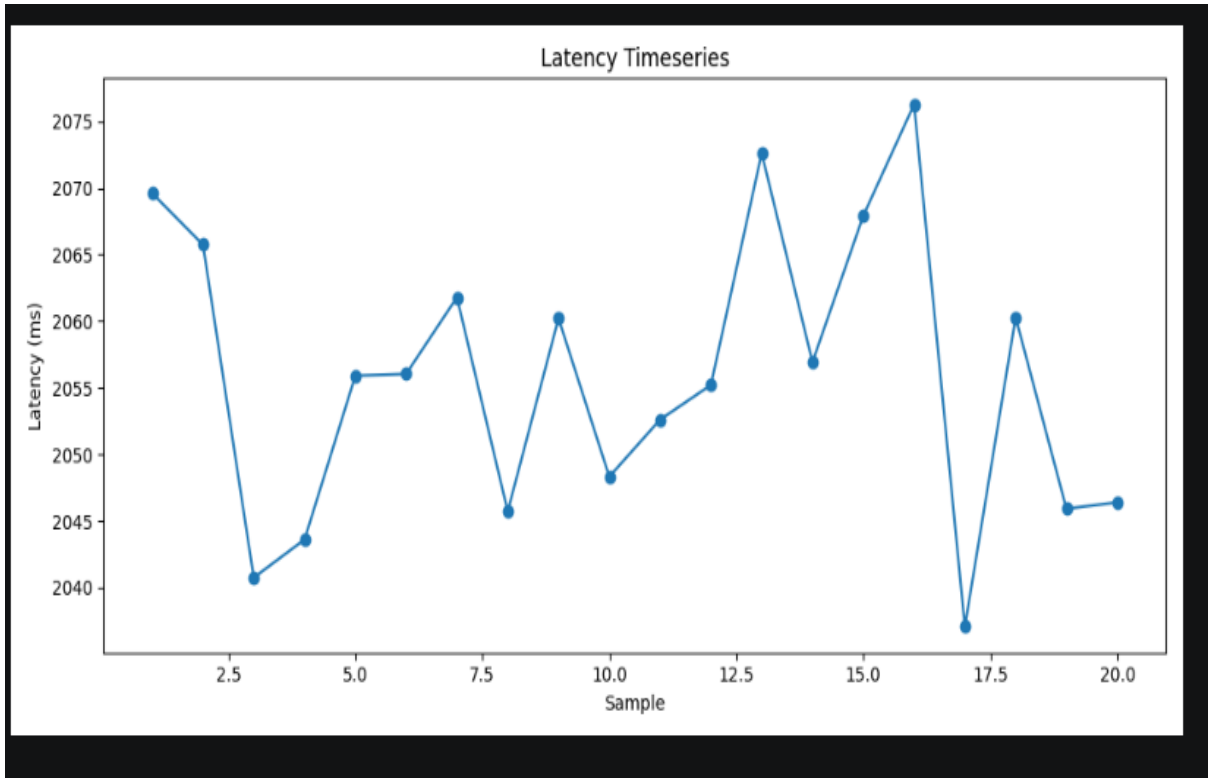




- GitHub Actions workflow
- Automatically:
- retrains models
- publishes new versions
- Supports continuous updates

SLIDE 10 — Monitoring

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> Get-Content artifacts\monitoring_report.json
{
  "total_samples": 20,
  "successful_samples": 20,
  "availability": 1.0,
  "error_rate": 0.0,
  "p95_latency_ms": 2072.61,
  "metrics_endpoint": "http://localhost:8000/metrics",
  "alert_policy": {
    "latency": "alert if p95_latency_ms > 1000",
    "errors": "alert if error_rate > 0.30",
    "availability": "alert if availability < 0.70"
  }
}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```



- Tracks system performance
- Metrics include:
- latency
- error rate
- availability
- Helps detect issues over time

SLIDE 11 — Availability Results

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> Get-Content artifacts\availability_report.json
{
  "availability": 1.0,
  "threshold": 0.7,
  "meets_requirement": true,
  "formula": "successful_samples / total_samples",
  "note": "This was measured over the local demo observation window. The same formula applies to the required 7-day availability win
dow."
}
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

- Availability = successful / total requests
- Example:
- 20 / 20 = 100%
- Exceeds required threshold (70%)

SLIDE 12 — Kafka Pipeline

```
PS C:\Users\mdzie\Desktop\MILESTONE-4> docker run --rm --network=milestone-4_default edenhill/kcat:1.7.1 -b kafka:29092 -L
Metadata for all topics (from broker 1: kafka:29092/1):
 1 brokers:
  broker 1 at kafka:29092 (controller)
 2 topics:
  topic "team.reco_responses" with 1 partitions:
    partition 0, leader 1, replicas: 1, isrs: 1
  topic "team.reco_requests" with 1 partitions:
    partition 0, leader 1, replicas: 1, isrs: 1
PS C:\Users\mdzie\Desktop\MILESTONE-4>
```

- Topics:
- reco_requests
- reco_responses
- Simulates real-time data flow
- Mimics production event-driven systems

SLIDE 13 — Fairness

•Issue: popularity bias
•Effects:
•same items recommended often
•low diversity
•Feedback loops:
•popularity loop
•tail starvation

SLIDE 14 — Security

- Risks:
- API abuse
- data poisoning
- message flooding
- Mitigations:
- schema validation
- logging + monitoring
- drift detection

SLIDE 15 — Conclusion

- Fully deployed ML system
- Supports:
- A/B testing
- model versioning
- monitoring
- Demonstrates real-world pipeline